Who actually decides?

Measuring voting-power concentration in Compound governance

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Token voting looks decentralised

- DAOs govern protocols holding billions of dollars by token-weighted on-chain votes.
- Compound: anyone holding COMP can vote, directly or by delegating.
- The usual decentralisation metrics are **token share** and the Gini coefficient.

But token share is not decision power.

- A 49% holder facing one 51% holder has *no* power; facing 51 holders of 1% each, it has almost all.
- Power depends on *who else* holds weight, the *quota*, and *delegation*.

Question: how far apart are nominal share and effective power in a real DAO, and how often did a handful of addresses decide outcomes?

Why it matters

Compound's on-chain votes decide its interest-rate and collateral parameters, which assets it lists, and payments from its treasury (contributor grants, security audits). Whoever holds decisive power controls all of that.

- **Delegators:** see whose vote actually decides outcomes before handing over their tokens.
- **Protocol designers:** test quorum, thresholds or vote caps on real data before changing the rules.
- **Security:** a small flip set means few addresses to bribe, buy or compromise to capture a vote.
- **Claims of decentralisation** (to users, investors, regulators) need a measure of power, not a count of holders.
- **Other DAOs:** Uniswap and many others run forks of the same governor, so the method carries over.

Existing approaches and their limitations

- **Token-share metrics** (Gini, Nakamoto coefficient): measure who *holds* votes, not who can *decide*. A 50% holder can have 60–67% of the power, and unequal holders can have equal power.
- **Fritsch, Müller & Wattenhofer (2022):** Compound, Uniswap, ENS, data to March 2022 via The Graph. Their "potential power" asks whether one delegate's vote could flip the *one vote that happened*, not across all coalitions. Address clustering is "not bullet-proof".
- **Messias et al. (2023):** Compound and Uniswap to August 2024; concentration (10 voters hold 50.53% of Compound's voting power) and voting costs. No power indices.

Gap: nobody computes exact Banzhaf and Shapley–Shubik indices over the delegation-resolved weights, for every proposal, with the weights checked against the chain. That is what we built.

What we did

1. Rebuilt **every** Compound GovernorBravo proposal (351) and vote (14,396) from raw Ethereum logs.
2. Reconstructed each delegate's voting weight at each proposal's snapshot by **replaying all 2.33M COMP transfers and 16,038 delegation events**.
3. Implemented delegation resolution, **Banzhaf** and **Shapley–Shubik** indices by dynamic programming ourselves.
4. Compared token share against both indices, and the indices against what actually happened in each vote.

Weighted voting games

- Players $N = \{1, \ldots, n\}$ with weights $w_i \geq 0$ and a quota q.
- A coalition S **wins** iff $w(S) = \sum_{i \in S} w_i \geq q$. Written $[q;\, w_1, \ldots, w_n]$.
- Player i **swings** for $S \subseteq N \setminus \{i\}$ if S loses but $S \cup \{i\}$ wins:

$$q - w_i \;\leq\; w(S) \;<\; q.$$

Example $[51;\, 50, 30, 20]$: winning coalitions $\{1,2\}, \{1,3\}, \{1,2,3\}$.
Player 1 holds 50% of the weight but is in every winning coalition.

Two power indices

Banzhaf: how often is i a swing, over all coalitions of the others?

$$\psi_i = \frac{\#\{S \subseteq N \setminus \{i\} : q - w_i \le w(S) < q\}}{2^{n-1}}, \qquad \beta_i = \frac{\psi_i}{\sum_j \psi_j}.$$

Shapley–Shubik: in a random ordering of players, how often is i the one whose arrival first reaches q?

$$\varphi_i = \sum_{\substack{S \subseteq N \setminus \{i\} \\ q - w_i \le w(S) < q}} \frac{|S|!\,(n-1-|S|)!}{n!}.$$

$[51; 50, 30, 20]$	player 1	player 2	player 3
token share	0.50	0.30	0.20
Banzhaf β	0.60	0.20	0.20
Shapley–Shubik φ	0.667	0.167	0.167

Delegation in COMP

- A holder's tokens vote only if it **delegates**, to itself or to another address.
- Delegation is **one-hop**: if $A \to B$ and $B \to C$, then A's tokens count for B, and B's own tokens for C.
- Undelegating ($\to$ `0x0`) removes the votes; undelegated tokens count for nobody.
- Voting weight at a proposal $= \sum$ balances delegated to you, at its `startBlock`.

Our resolver implements this rule, plus a *transitive* variant for sensitivity:

- follow $A \to B \to C \to \cdots$ to a self-delegated or non-delegating address;
- **cycle rule:** addresses on a cycle are treated as self-delegated; a chain entering a cycle stops at the first cycle address.

Pipeline

Step	What happens
1. Collect	Download every proposal, vote, COMP transfer and delegation from Ethereum (COMP's launch to 2025-03-08), saved locally.
2. Reconstruct	Replay transfers and delegations in order to get each delegate's voting weight when each vote opens.
3. Model	Each proposal becomes a weighted voting game: delegates are the players, voting weights their weights.
4. Measure	Banzhaf and Shapley–Shubik power of every delegate; the smallest group that could have reversed each outcome.
5. Compare	Power vs token share, and vs what actually happened: every table, figure and number in this talk.

- **Scope:** GovernorBravo proposals 43–393, 2021-04-14 to 2025-01-30 (281 executed, 33 defeated, 37 canceled).
- **Players:** every address holding voting weight when the vote opens (median 4,912).
- **Passing rule:** for $>$ against and for $\geq$ 400k COMP $\Rightarrow$ quota $q = \max(\lfloor W/2 \rfloor + 1,\ Q)$.

The data is exact

We never trust a third-party API for weights; instead we check our reconstruction against the chain itself.

Check	Mismatches
Our delegate weights vs on-chain checkpoints (all delegates, 351 snapshots)	**0**
Our weight vs the weight of each of the 14,396 `VoteCast` events	**0**
Our for/against/abstain tallies vs `proposals(id)` (max difference, COMP)	**2e-10**
Our outcome label vs `state(id)`	**0**

This validates the delegation resolver and the snapshot rule (events in block $\leq$ `startBlock`) end to end. The index code is checked by 280 tests, including against brute-force enumeration (backup slide).

Computing the indices: dynamic programming

Banzhaf. $p[x] = \Pr(w(S) = x)$ for S a random subset; add a player of weight v:

$$p'[x] = \tfrac{1}{2}\big(p[x] + p[x - v]\big), \qquad x < q.$$

Shapley–Shubik. $P[k, x] = \Pr(w(S) = x)$ for S a random k-subset of the m players so far:

$$P'[k, x] = \frac{(m + 1 - k)\, P[k, x] + k\, P[k - 1, x - v]}{m + 1},$$

$$\varphi_i = \frac{1}{n} \sum_{k=0}^{n-1} \Pr_k\big(q - w_i \le w(S) < q\big).$$

- Convex combinations: no overflow, even for thousands of players (raw counts reach $\binom{n}{n/2}$).
- Leave-one-out by **divide and conquer** over distinct weights: only additions, no unstable "removal".
- Weights rounded to $W/20000$; dummies carry $\le 0.1\%$ of weight. Changing the resolution moves any index by ≤ 0.00016.

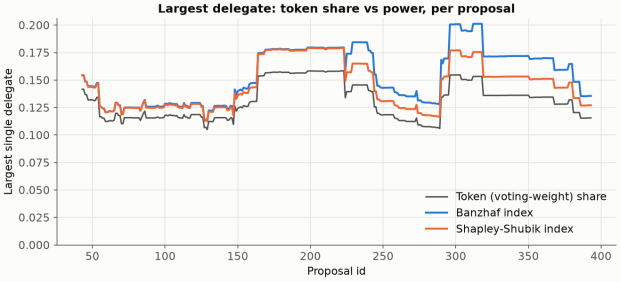
A few delegates hold most of the weight

Median over 351 proposals:

	largest delegate
token share	13.1%
Banzhaf	14.9%
Shapley–Shubik	14.3%

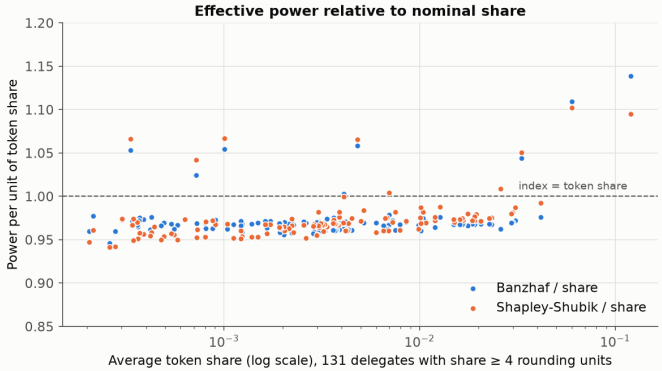
Top 5: 38.2% of weight,
39.6% / 40.6% of power.

Of 4,912 players, only ∼76 are not dummies.



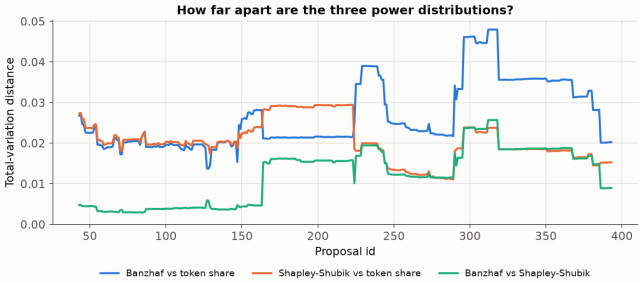
Nominal vs effective: the largest are amplified

- The 2 largest delegates get 1.09–1.14× power per unit of tokens.
- The other 129 get a median 0.97×; 95.3% of them are below 1.
- Largest single gap: +4.8 points (Banzhaf), +2.2 points (Shapley–Shubik), both for the top delegate.



Banzhaf vs Shapley–Shubik: same order, different sizes

- Spearman rank correlation is 1.000 in *every* proposal: a theorem, since a heavier player is never less powerful under either index.
- The real difference is in size: total-variation distance median 0.015 (max 0.026).
- From proposal 224 (2024-03-13) on, Banzhaf gives the top delegate $\geq$0.8 points more.

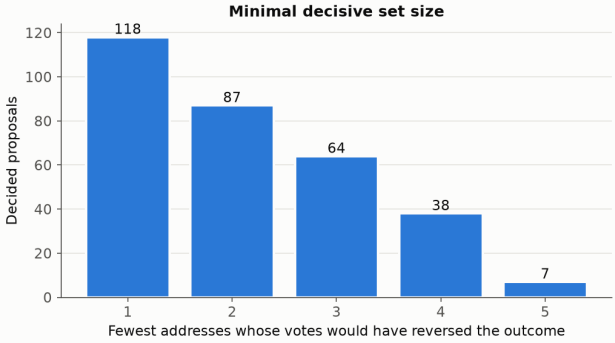


From power to outcomes: how many addresses decided?

Flip set: players who, changing only their own votes, reverse the result (with quorum).

Of 314 decided proposals:

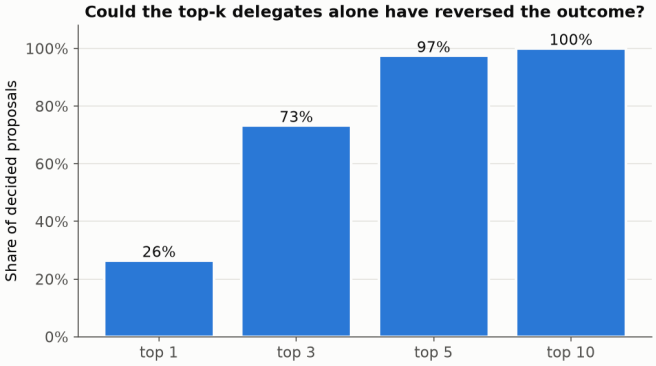
- **118** could be reversed by **one** address
- median minimal set: 2; never more than 5
- 28 of 33 defeated proposals: one large non-voter could have passed them



Could the top-k have reversed it alone?

k	share of outcomes
1	26.4%
3	73.2%
5	97.5%
10	100.0%

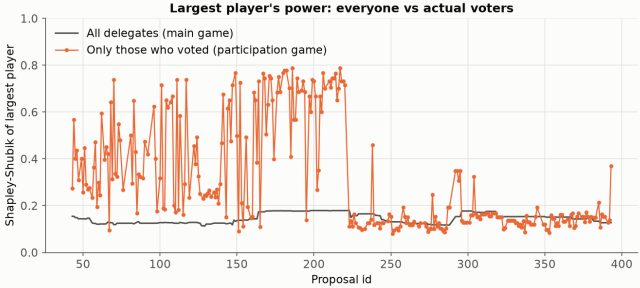
Median turnout: 25.5% of delegated weight. Low turnout is what makes a few large delegates decisive.



Beyond the base game: power among those who actually vote

Players = the **for/against voters** only; same quota rule on votes cast.

- Quorum (400k COMP) now binds in 259 of 314 decided proposals.
- In 23, the voters could not reach quorum at all.
- Median: 21 voters; the largest holds 17.6% of cast votes but 18.3% Shapley–Shubik power.
- Nakamoto (Shapley–Shubik) drops from 9 to 4.



Agreement with prior work

We recomputed the published Compound metrics on our own data.

Source	Metric	Published	Ours
Fritsch, Müller & Wattenhofer (2022)	Nakamoto coefficient of delegates, block 14.3M (2022-03-01)	8	**8**
Fritsch, Müller & Wattenhofer (2022)	Gini coefficient of delegates, block 14.3M (2022-03-01)	0.987	**0.986**
Messias et al. (2023)	mean voters needed for $\geq$50% of votes cast	3.18	**3.55**

- **Different periods:** Fritsch et al. data to 2022-03-01 (we compare at our nearest snapshot, block 14,252,261); Messias et al. to 2024-08-19, including Governor Alpha; ours 2021-04-14 to 2025-01-30 (314 decided Bravo proposals).
- **What we add:** neither paper computes Banzhaf or Shapley–Shubik indices. Fritsch et al. use "could one delegate flip this vote"; we compute both indices exactly over the delegation-resolved game, for every proposal.

Takeaways

1. **Concentration is real, and slightly worse than token share suggests.** The top delegate's power exceeds its share under both indices in 100.0% of proposals; the median Nakamoto coefficient falls from 10 (weight) to 9 (Shapley–Shubik).
2. **The two indices agree on who matters** (by theorem) and differ by only $\sim$0.015 in total variation, but Banzhaf amplifies the top delegate more.
3. **Among actual voters it is starker:** quorum binds in 259 of 314 proposals and 4 voters hold most of the power.
4. **Realised outcomes are fragile.** The top 5 delegates could have reversed 97.5% of all decided proposals; a single address, 118 of 314.

Limitations

- **Snapshot choice:** weights at `startBlock`, exactly as the protocol counts; delegation can change during voting but cannot affect votes.
- **Delegation rule:** one-hop, as in COMP. A transitive reading moves at most 5.3% of weight in any proposal.
- **Addresses $\neq$ people:** one entity may control many addresses, which would make concentration *higher*. Fritsch et al. cluster addresses by transfers but call it "not bullet-proof"; we do not cluster.
- **Model assumptions:** the index game assumes all players vote and are not strategic (no coordination). Real turnout is 25.5%; the participation game (slide on actual voters) relaxes the all-vote assumption.
- **Rounding:** integer weights at $W/20000$; effect ≤ 0.00016 on any index.
- **Scope:** one DAO and one governor contract (GovernorBravo, 2021-04-14 to 2025-01-30). Fritsch et al.'s data stop at 2022-03-01.

Future directions

- **Turnout models:** between our two games, weight players by their historical participation.
- **Entity clustering:** merge addresses controlled by the same organisation.
- **Second DAO:** Uniswap uses the same token/governor design, so the pipeline transfers with a config change.
- **Off-chain voting** (Snapshot) and liquid-delegation designs where chains really are transitive.
- **Mechanism design:** how quorum or vote caps would change the indices.

References: Fritsch, Müller & Wattenhofer, *Analyzing Voting Power in Decentralized Governance: Who controls DAOs?*, Blockchain: Research and Applications 5(3), 2024, doi:10.1016/j.bcra.2024.100208 (arXiv:2204.01176).

Messias et al., *Understanding Blockchain Governance: Analyzing Decentralized Voting to Amend DeFi Smart Contracts*, arXiv:2305.17655.

Thank you

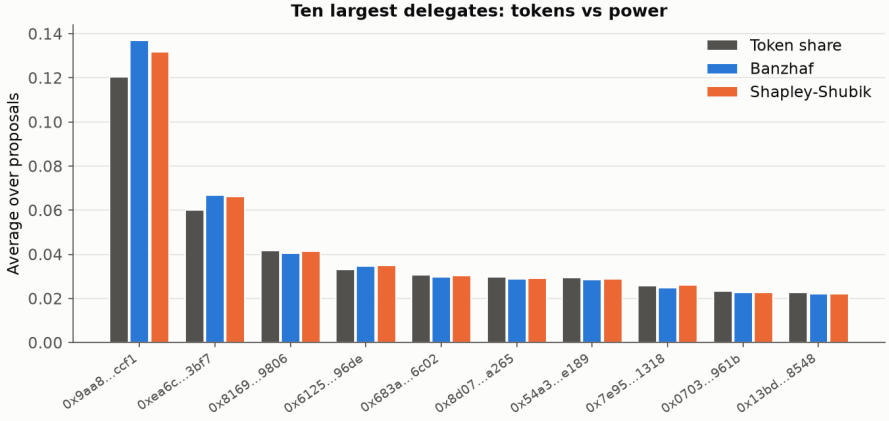
Questions?

Code, data pipeline and all results: `README.md` section 10 reproduces every number.

Backup slides

For questions

Ten largest delegates



Averaged over all proposals. Addresses are shown, not identities: one entity may control several.

Their power measure vs our indices

Potential power (Fritsch et al., Def. 4.1): could this voter, by switching its vote, have flipped the result? **Exercised power** (Def. 4.2): did removing its vote flip it?

On our data (Compound rule, with quorum)

- Only 34 of 2,338 voters *ever* had potential power, in 96 of 314 proposals.
- One delegate could have flipped 57 of the 84 proposals it voted on.
- Sanity check: every such proposal also has a one-address flip set in our analysis (asserted in code).

Why we need full indices

- Potential power checks a swing at *one* coalition, the one that happened. Banzhaf averages swings over *all* 2^{n-1} coalitions; Shapley–Shubik over all orderings.
- It is 0 for most voters in most votes, so it cannot rank them; our indices can.
- Yet they agree: potential-power rate vs our Shapley–Shubik, Spearman 0.70 over 552 voters with $\geq$5 votes.

Tests (280, all passing)

Indices

- known example $[51; 50, 30, 20]$ to 10^{-9}
- dummy players get 0
- equal weights $\Rightarrow$ equal indices
- Shapley–Shubik sums to 1
- DP = brute force on 40 random games with $n \leq 12$
- DP = permutation definition ($n \leq 7$)

`pytest` runs in about 2 seconds.

Delegation & outcomes

- chains, cycles, self-delegation, undelegation, re-delegation
- same-block ordering by log index
- snapshot unaffected by later events
- one-pass replay = direct recomputation
- minimal flip set = brute force on 200 random votes